



Cite this: *Soft Matter*, 2024,
20, 4389

Received 17th February 2024,
Accepted 3rd May 2024

DOI: 10.1039/d4sm00221k

rsc.li/soft-matter-journal

Surface mobility gradient and emergent facilitation in glassy films

Qiang Zhai,^a Xin-Yuan Gao,^b Chun-Shing Lee,^c Chin-Yuan Ong,^d Ke Yan,^e Hai-Yao Deng,^f Sen Yang^g and Chi-Hang Lam^g

Confining glassy polymers into films can substantially modify their local and film-averaged properties. We present a lattice model of film geometry with void-mediated facilitation behaviors but free from any elasticity effect. We analyze the spatially varying viscosity to delineate the transport properties of glassy films. The film mobility measurements reported by Yang *et al.*, *Science*, 2010, **328**, 1676 are successfully reproduced. The flow exhibits a crossover from a simple viscous flow to a surface-dominated regime as the temperature decreases. The propagation of a highly mobile front induced by the free surface is visualized in real space. Our approach provides a microscopic treatment of the observed glassy phenomena.

1 Introduction

Glassy films have drawn substantial research interest.^{1–3} The presence of interfaces either with air or a substrate was reported to alter the glass transition temperature T_g ,^{4–8} film viscosity^{9–11} and relaxation time.^{12,13} In general, a free surface tends to facilitate the relaxation while a strongly attractive substrate or adsorbed layer tends to slow down the dynamics.¹⁴ Early studies suggest that interfaces induce a gradient in T_g ¹⁵ and the α -relaxation time,¹⁶ both being important characteristics of structural relaxation in glasses. Besides, the observed dynamical gradients penetrate much deeper than the particle density gradients.^{16,17} Impacts on confinement effects by polymer chain connectivity¹⁸ and a possible non-equilibrium nature of film samples¹⁹ are also interesting questions.

The reduction of T_g has been interpreted in various theoretical postulations.^{4,20–25} In particular, a two-layer model, namely a highly mobile non-glassy layer atop a bulk-like glassy inner layer, is used to analyze the film mobility inferred from surface roughening experiments.⁹ The two-layer model serves as a simplified approach to account for a smooth gradient of T_g and the relaxation

time.^{17,26–29} Note that a decoupling^{30,31} between the segmental mobility and T_g reduction was also reported for polystyrene films³² and polymer nanospheres.³³ For instance, the relaxation time demonstrates a lack of film-thickness dependence in dielectric measurements,³² despite a thickness-dependent T_g reduction.

Lattice models have been instrumental in understanding glassy phenomena.³⁴ Compared with molecular dynamics (MD) simulations, lattice models are more simplified and aim to capture the essential features of glassy systems. Such simplicity often allows detailed analysis and provides insights into the glass transition. Recently, the dynamical facilitation picture of glass,³⁵ originally motivated by lattice models,^{36,37} has received a surge of interest,^{38–40} with much effort focusing on long-range elasticity as a possible mediator of facilitation.^{39,40}

Here we study the dynamics of glassy films using a lattice model with emergent facilitation but free from long-range elasticity. We reproduce a spatial gradient in the relaxation time, which lends support to the two-layer model. We also demonstrate a void-mediated facilitation mechanism. Our study is based on a distinguishable-particle lattice model (DPLM)⁴¹ free from long-range elastic interactions. It has been shown to be capable of reproducing a plethora of glassy phenomena with little modification of the model, including Kovacs paradox⁴² and effect,⁴³ fragility,⁴⁴ heat capacity overshoot,⁴⁵ two-level systems,⁴⁶ diffusion coefficient power-laws⁴⁷ and Kauzmann's paradox.⁴⁸ Dynamical facilitation behaviors resembling those from MD simulations are also observed, without invoking elasticity, facilitation rules, or kinetic constraints.

2 Film lattice model

A schematic drawing of the lattice model is displayed in Fig. 1. A DPLM film of thickness h in two dimensions (2D) comprises N distinguishable particles – each of its own type – sitting on a

^a School of Physics, MOE Key Laboratory for Nonequilibrium Synthesis and Modulation of Condensed Matter, Xi'an, Shaanxi, 710049, China. E-mail: yangsen@mail.xjtu.edu.cn

^b Department of Physics, The Chinese University of Hong Kong, Shatin, New Territories, Hong Kong, China

^c School of Science, Harbin Institute of Technology (Shenzhen), Shenzhen, 518055, China

^d School of Physics, Yale University, New Haven, Connecticut, 06520, USA

^e School of Mechanical Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi, 710049, China. E-mail: yanke@mail.xjtu.edu.cn

^f School of Physics and Astronomy, Cardiff University, 5 The Parade, Cardiff, CF24 3AA, Wales, UK. E-mail: dengh4@cardiff.ac.uk

^g Department of Applied Physics, Hong Kong Polytechnic University, Hong Kong, China. E-mail: C.H.Lam@polyu.edu.hk

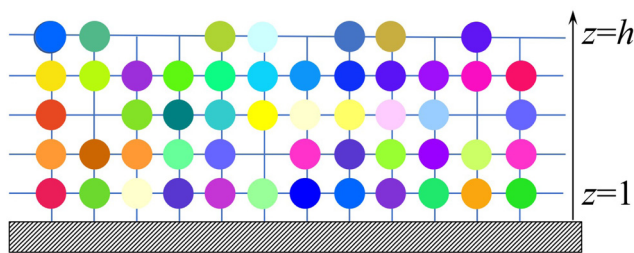


Fig. 1 A schematic drawing of the 2D lattice model for a supported film of thickness h , in which all the particles are distinguishable.

square lattice terminated along the z -axis but with periodic boundary conditions applied along the x -axis. We have recently shown that the DPLM behaves qualitatively similarly in two and three dimensions.⁴⁹ We let the film be supported on a substrate on one termination (bottom layer, designated by $z = 1$) while the other termination is a free surface (top layer, $z = h$). The energy of the film is given by

$$E = \sum_{\langle i,j \rangle} V_{s_i s_j} n_i n_j + \varepsilon_{\text{top}} \sum_{i: z_i=h} n_i + \varepsilon_{\text{bot}} \sum_{i: z_i=1} n_i, \quad (1)$$

Here the first summation is over pairs of adjacent sites as indicated by $\langle i,j \rangle$, n_i is the particle occupation at site i , which equals either one if a particle is present or zero if a void (the absence of particle) is present, $V_{s_i s_j}$ represents the interaction between a particle at site i of type s_i and a particle at an adjacent site j of type s_j , $\varepsilon_{\text{top}} = 1.124$ accounts for the extra energy of a particle at the free surface while $\varepsilon_{\text{bot}} = -0.5$ accounts for substrate effects, and $z_i = 1, 2, \dots, h$ is the z -coordinate of site i . In our system, there are in total N particle types so $s_i = 1, 2, \dots, N$. The interaction $V_{s_i s_j}$ is sampled before the commencement of the simulation from the *a priori* probability distribution,⁴⁴

$$g(V) = G_0/(V_1 - V_0) + (1 - G_0)\delta(V - V_1), \quad (2)$$

where $V_0 = -0.5$, $V_1 = 0.5$ and $G_0 = 0.7$. The equilibrium void-induced dynamics⁵⁰ is simulated by the Metropolis algorithm, with the acceptance rate,

$$w = w_0 \exp[-\Delta E \Theta(\Delta E)]/k_B T, \quad (3)$$

where ΔE is the change in the system energy E defined by eqn (1) after a particle hop attempt. $\Theta(x) = 1$ if $x > 0$, or $\Theta(x) = 0$ otherwise. The attempt frequency $w_0 = 10^6$ and the Boltzmann constant k_B is set to 1 for simplicity.

We focus on equilibrium studies. To prepare a sample at the working temperature T , a swap algorithm (swapping among all the particles and voids⁴⁷) is used to speed up the equilibration process. This process continues until the energy and void density are stabilized during more than 10^{10} extra swap Monte Carlo attempts. All results reported below are based on fully equilibrated samples.

The particle density measured from equilibrium simulations is displayed in Fig. 2. Across the temperature range we have studied, the surface void density ϕ_v^s is kept at 80% by fine tuning N . This also determines the void density ϕ_v ($\ll \phi_v^s$) in the bulk-like inner region which becomes temperature dependent

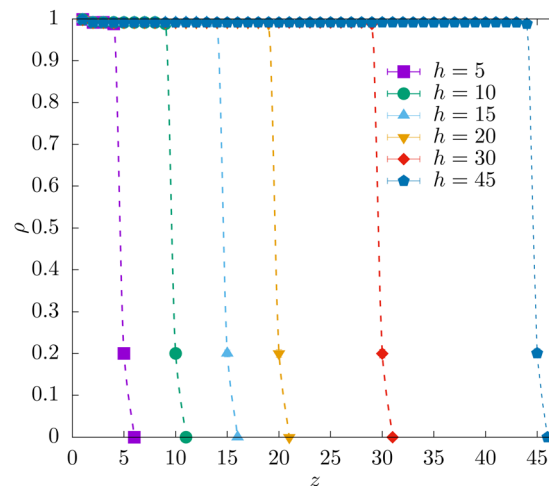


Fig. 2 The layer-resolved particle density obtained after sample equilibration at $T = 0.22$. The particle density in the inner region ($2 \leq z \leq h - 2$) is nearly constant with a sharp drop at the surface.

but film thickness independent, as shown in Fig. 3. Only a single topmost layer at $z = h$ has a significantly reduced particle density, modeling a very sharp interface with a high local particle mobility analogous to a free surface. The particle densities are nearly a constant independent of z except very close to the free surface and the substrate. Yet, the gradient of the relaxation time covers a thicker region than the density gradient.

3 Results and discussion

The depth-resolved α -relaxation time $\tau_\alpha(z)$ is extracted from an overlap function $q(t, z)$ obtained under thermal equilibrium. The overlap function $q(t, z)$ is defined as the probability that a particle residing initially in layer z experiences no net movement after duration t . It has been successfully used to quantify the structural relaxation in bulk DPLM.⁴² In Fig. 4(a), $q(t, z)$ is shown

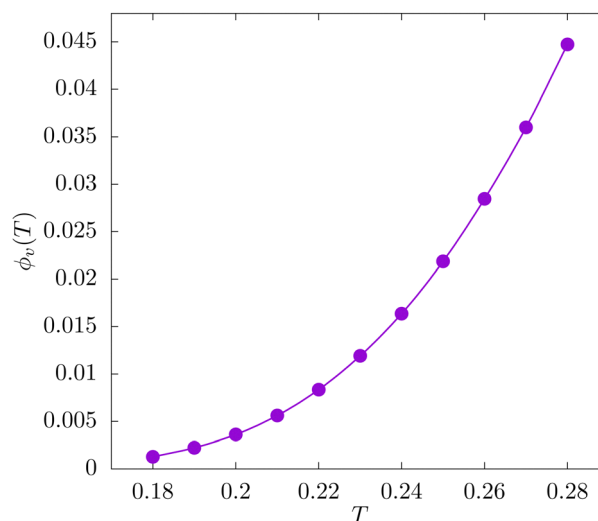


Fig. 3 Plot of the void density in the inner region of the film away from the free surface and the substrate as a function of T .

at various z for a film of thickness $h = 45$ at a low temperature $T = 0.20$. The time window spans over ten orders. From its value of unity at $t = 0$, $q(t, z)$ decays as t increases due to particle motions. It is seen that for the top layer ($z = 45$) the decay is fast, but it becomes slow for deeper layers. Then, $\tau_\alpha(z)$ is defined as the time when $q(t, z)$ drops to $1/e$ with e being the Euler constant, i.e. $q(\tau_\alpha(z), z) = 1/e$. The extracted local relaxation rate, $1/\tau_\alpha(z)$, is shown at various temperatures in Fig. 4(b). Here one notes that $1/\tau_\alpha(z)$ decays toward a bulk value as z moves into deeper layers. The thickness of the surface affected region can extend beyond ten layers at $T = 0.19$.

Now we employ the as-obtained $\tau_\alpha(z)$ to analyze the film mobility M . Our strategy is based on the notion that the local viscosity $\eta(z)$ of a film can be approximated from $\tau_\alpha(z)$, i.e. $\eta(z) = G\tau_\alpha(z)$, known as Maxwell's relation. Here G is the shear modulus of the film, which is assumed with negligible temperature dependence. The mobility M measures the amount of flux that can be carried by the film under a unit pressure gradient parallel to the film.⁹

Specifically, we apply the Navier-Stokes equation and the lubrication approximation to film flow along the x -direction under no-slip boundary conditions at the bottom and free boundary conditions at the top,

$$\begin{aligned}\partial_z[\eta(z)\partial_z v_x(z)] - \partial_x P &= 0, \\ v_x(z=0) &= 0, \\ \partial_z[v_x(z)]|_{z=h} &= 0,\end{aligned}\quad (4)$$

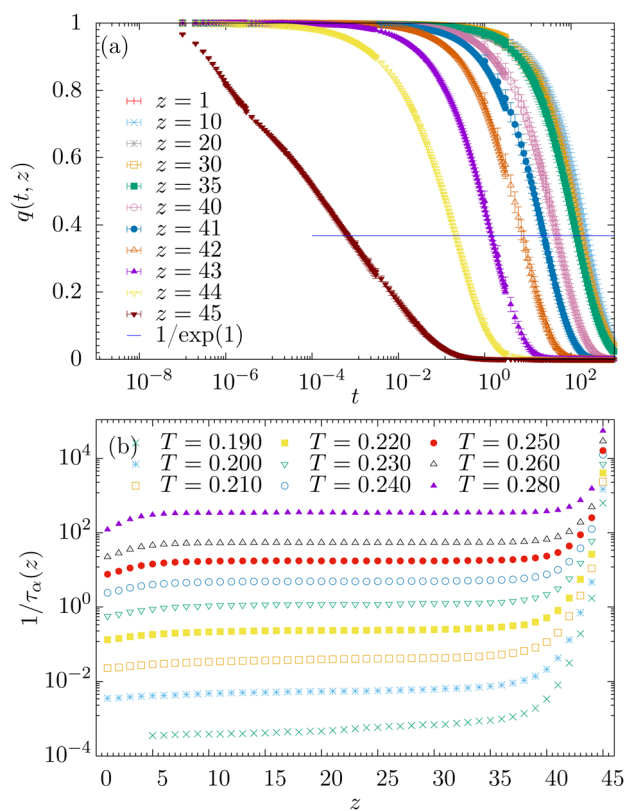


Fig. 4 (a) Layer-resolved overlap function $q(t, z)$ for film thickness $h = 45$ at $T = 0.20$. (b) Local relaxation rate $1/\tau_\alpha(z)$ versus distance z from the substrate at various temperatures.

where P is the pressure, $\eta(z)$ is the layer-dependent local viscosity and $v_x(z)$ is the velocity in the x direction. After solving eqn (4), we find,

$$v_x(z) = -\partial_x P \int_0^z \frac{(h-z')}{\eta(z')} dz'. \quad (5)$$

The total flux is,

$$J = \int_0^h dz v_x(z) = -\partial_x P \int_0^h dz \int_0^z \frac{(h-z')}{\eta(z')} dz'. \quad (6)$$

The mobility of the film is defined as,⁹

$$M = -J/\nabla P. \quad (7)$$

We substitute eqn (6) into eqn (7) and integrate by part, leading to

$$M = \int_0^h dz \frac{(h-z)^2}{\eta(z)}. \quad (8)$$

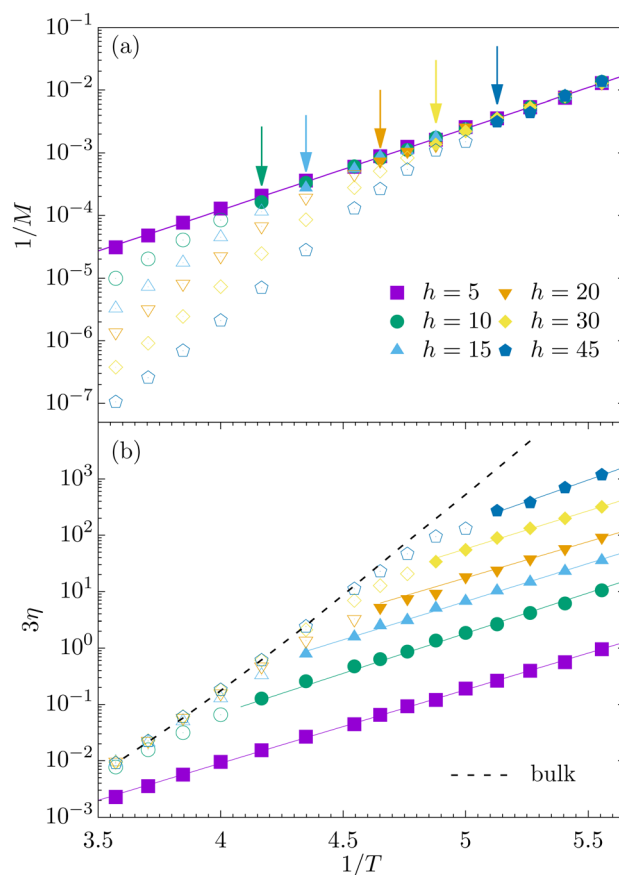


Fig. 5 (a) Film mobility M and (b) effective viscosity η versus inverse temperature $1/T$ for films of various thicknesses h . Data points at each thickness can be represented by either solid or open symbols to indicate surface or bulk-like flow respectively. Arrows show the crossover points between the flow regimes.

Finally, by approximating $\eta(z)$ with $G\tau_\alpha(z)$ and taking a shear modulus $G = 1$ for simplicity, we reach,

$$M = \int_0^h dz \frac{(h-z)^2}{\tau_\alpha(z)}. \quad (9)$$

The effective film viscosity η is then defined by $M = \eta^{-1} \int_0^h dz (h-z)^2$, which restores $3\eta = h^3/M$ for an open-channel viscous flow. In Fig. 5 is shown M and η against temperature for films of various thicknesses. As seen in panel (a), at low temperatures the mobility M of thinner films collapses, indicating the existence of a surface layer dominating the film transport capacity. The thickness of such a layer is at most five. In contrast, from panel (b), the effective viscosity η of thicker films converges to the bulk value at high temperatures, signifying bulk-like whole-film flow. All of these features well agree with experimental observations on short-chain polystyrene films.⁹

To better understand the contribution of the local relaxation time $\tau_\alpha(z)$ to the mobility M , we plot the integrand, $(h-z)^2/\tau_\alpha(z)$, in Fig. 6. At high T , the relaxation time varies spatially only by less than two orders of magnitude, and the geometric factor $(h-z)^2$ dominates. In contrast, at low T , the spatial variation of the relaxation time becomes much more pronounced and consequently surface flow dominates the total flow as shown Fig. 6.

We now illustrate particle and void dynamics in real space. Fig. 7 shows snapshots of a film at $T = 0.19$ at time t with particles colored according to their displacement $|r_i(t) - r_i(0)|$ where $r_i(t)$ denotes position of particle i at time t . The system has been fully equilibrated at time $t = 0$ so that the monitored particle motions are equilibrium-dynamics. As seen, mobile domains (light blue to red) gradually expand and invade unrelaxed regions (deep blue), demonstrating dynamical facilitation resembling observations in MD simulations.^{38,51,52} In the deeper bulk-like inner region, mobile domains vary greatly in sizes, reproducing dynamical heterogeneity characteristic of glass.⁵³ Facilitation among particles in DPLM are void mediated. While smaller domains are induced by single voids, the larger ones are initiated by multiple coupled voids which facilitate each other's motions.⁴¹

Moreover, the free surface initiates a rather smooth mobile front (red) in Fig. 7. The large particle displacements there can easily be seen from similar snapshots as resulting from a continuous flux of isolated voids randomly emerging from and vanishing into the free surface, in contrast to their bulk counterparts which are mostly trapped. Motions of voids in DPLM are sub-diffusive at low temperature because of the random energy landscape quenched in the configuration space.^{54,55} This may explain the subdiffusive growth of the mobile front, as observed in MD simulations.⁵²

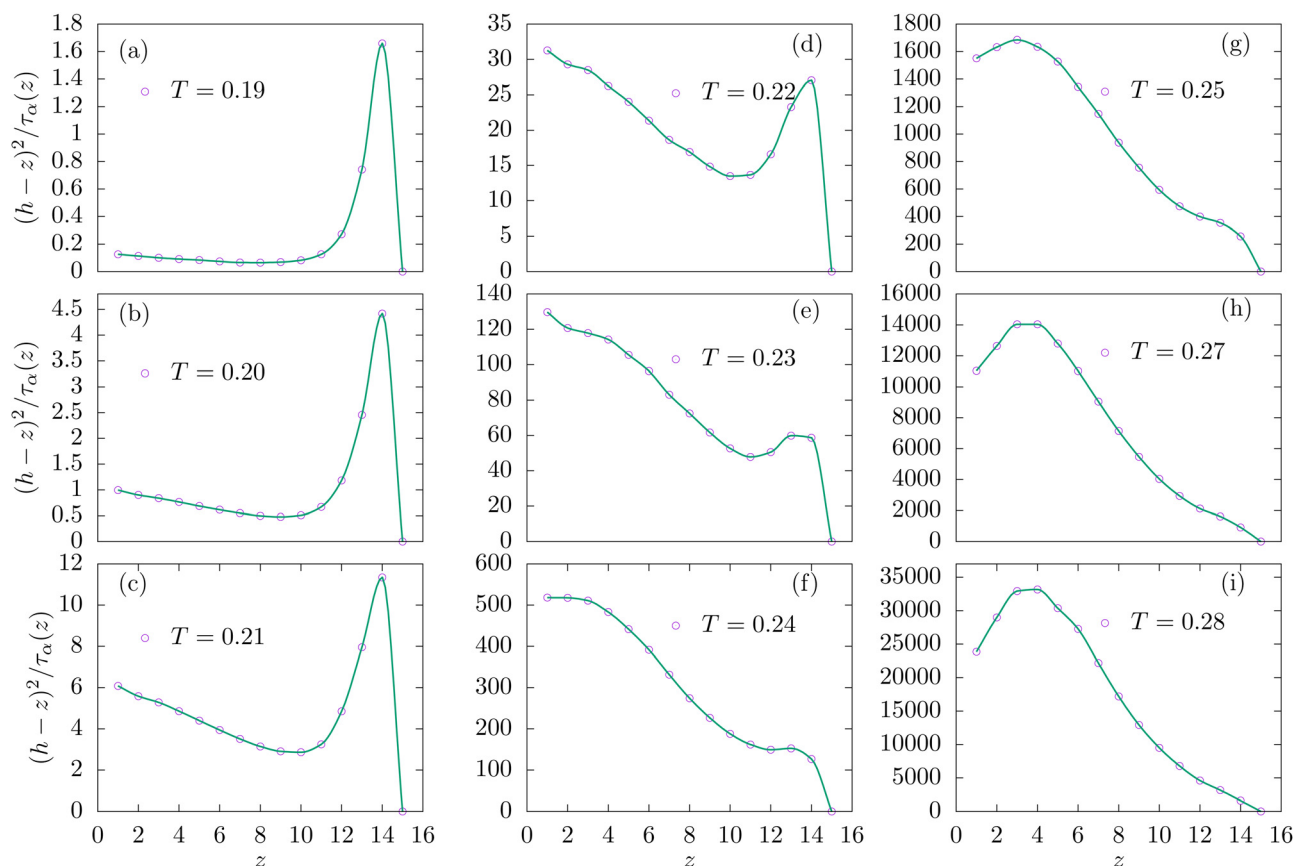


Fig. 6 The integrand in eqn (9) is plotted for the $h = 15$ film at different temperatures. Importance of surface relaxation increases dramatically at $T \leq 0.22$, signifying surface transport.

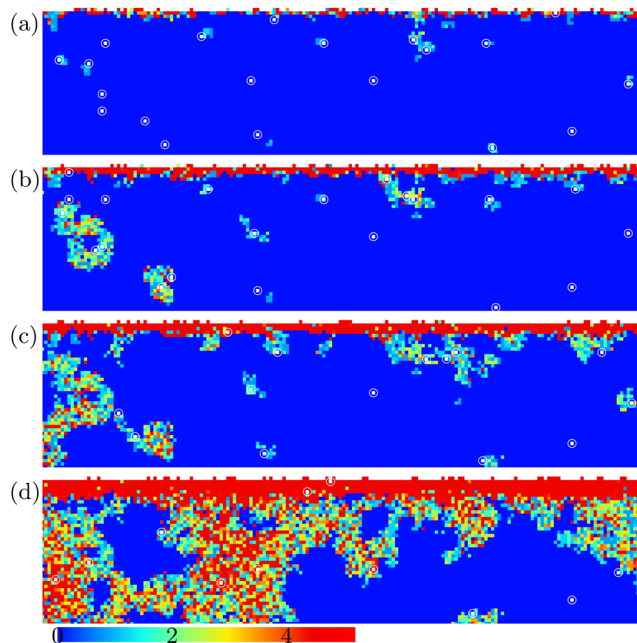


Fig. 7 Snapshots showing the temporal evolution of particle displacement $|r_i(t) - r_i(0)|$ for film thickness $h = 45$ at $T = 0.19$ and time $t = 2$ (a), 20 (b), 72 (c) and $\tau_z^{\text{bulk}} = 1550$ (d). White squares, highlighted by circles, denote voids.

Finally, we examine detailed structures of the mobile front in Fig. 7. The smooth main front discussed above is indeed punctuated by isolated plumes, resembling similar structures reported in previous MD simulations.¹⁷ These plumes are often seen to be induced by motions of coupled pairs of voids, as opposed to isolated voids. Despite being fewer in number, coupled voids are significantly more mobile due to facilitation⁴¹ and can penetrate deeper beneath the surface.

4 Conclusions

The DPLM encodes the rugged energy landscape typical of glasses through particle-type-dependent interactions. Compared with spin glasses,⁵⁶ where the disordered interaction is quenched in real space, the interaction in the DPLM is quenched in the configuration space. The void-induced dynamics has recently been evidenced in a colloidal glass former.⁵⁰

To apply findings on films to better understand glass in general as originally envisioned,⁴ a unified approach for both film and bulk geometries is highly desirable. Existing analytical theories on film dynamics^{23–25,57,58} inevitably require film-specific assumptions, e.g. a layer-by-layer facilitation-like process in,²⁵ and this complicates thorough testing of underlying concepts against both film and bulk phenomena. To this end, lattice models can provide an intermediate bridge between theories and MD simulations or experiments so that assumptions can be reliably tested. In this work, the DPLM algorithm adopted is identical to that in previous works on bulk^{44–46} with trivial differences only in the boundary layers (see eqn (1)). It provides a truly unified approach with implications generally applicable to both film and bulk.

An interesting feature demonstrated by our model is the decoupling between local density gradient and relaxation dynamics. In our case, the particle density is only reduced at the top-most layer, while in sharp contrast, the relaxation time gradient can span about ten layers at the lowest temperature that we have simulated. This is consistent with observations in MD simulations.^{16,17,59} The present work shows that DPLM can naturally reproduce the decoupling, without any additional assumption or modification. In particular, it is demonstrated explicitly that surface enhanced mobility at a given depth does not require a higher local density of voids. We believe that the mobility enhancement in our simulations is instead due to a breakdown of the anti-correlation of particle motions as suggested in ref. 17. This will be further studied in the future.

In conclusion, lattice model simulations in film geometry successfully reproduced confinement effects characteristic of glassy short-chain polymer or small-molecule films, namely the crossover from whole-film flow to surface flow and the reduced relaxation time deep into the film. Enhanced dynamics as facilitated by the free surface is visualized. Our results show that void-induced particle dynamics is able to account for these phenomena, without invoking long-range elasticity, sample non-equilibrium conditions or polymer chain connectivity.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

We thank Ophelia Tsui for helpful discussions. This work was supported by the China Postdoctoral Science Foundation (Grant No. 2022M722548), the Natural Science Foundation of Shaanxi Province (Grant No. 2023-JC-QN-0018), the Fundamental Research Funds for the Central Universities (Grant No. xzy012023044), the Hong Kong GRF (Grant No. 15303220), and the National Key Research and Development Program (Grant No. 2020YFB2007901 and No. 2022YFE0109500).

Notes and references

- 1 G. B. McKenna and S. L. Simon, *Macromolecules*, 2017, **50**, 6333.
- 2 C. B. Roth, *Chem. Soc. Rev.*, 2021, **50**, 8050.
- 3 K. S. Schweizer and D. S. Simmons, *J. Chem. Phys.*, 2019, **151**, 240901.
- 4 J. L. Keddie, R. A. L. Jones and R. A. Cory, *Eur. Phys. Lett.*, 1994, **27**, 59.
- 5 J. A. Forrest, K. Dalnoki-Veress, J. R. Stevens and J. R. Dutcher, *Phys. Rev. Lett.*, 1996, **77**, 4108.
- 6 J. E. Pye and C. B. Roth, *Phys. Rev. Lett.*, 2011, **107**, 235701.
- 7 O. Bäumchen, J. D. McGraw, J. A. Forrest and K. Dalnoki-Veress, *Phys. Rev. Lett.*, 2012, **109**, 055701.
- 8 R. Inoue, T. Kanaya, T. Yamada, K. Shibata and K. Fukao, *Phys. Rev. E*, 2018, **97**, 012501.

- 9 Z. Yang, Y. Fujii, F. K. Lee, C.-H. Lam and O. K. C. Tsui, *Science*, 2010, **328**, 1676.
- 10 C. M. Evans, S. Narayanan, Z. Jiang and J. M. Torkelson, *Phys. Rev. Lett.*, 2012, **109**, 038302.
- 11 Y. Chai, T. Salez, J. D. McGraw, M. Benzaquen, K. Dalnoki-Veress, E. Raphaël and J. A. Forrest, *Science*, 2014, **343**, 994.
- 12 Z. Fakhraai and J. A. Forrest, *Science*, 2008, **319**, 600.
- 13 B. Sun, W. Cao, Z. Wang, B. Sun and W. Wang, *Phys. Rev. B*, 2022, **105**, 014110.
- 14 J. A. Torres, P. F. Nealey and J. J. de Pablo, *Phys. Rev. Lett.*, 2000, **85**, 3221.
- 15 C. J. Ellison and J. M. Torkelson, *Nat. Mater.*, 2003, **2**, 695.
- 16 F. Varnik, J. Baschnagel and K. Binder, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2002, **65**, 021507.
- 17 C.-H. Lam, *J. Chem. Phys.*, 2018, **149**, 164909.
- 18 C. B. Roth, A. Pound, S. W. Kamp, C. A. Murray and J. R. Dutcher, *Eur. Phys. J. E: Soft Matter Biol. Phys.*, 2006, **20**, 441.
- 19 A. Panagopoulou and S. Napolitano, *Phys. Rev. Lett.*, 2017, **119**, 097801.
- 20 S. Herminghaus, *Eur. Phys. J. E: Soft Matter Biol. Phys.*, 2002, **8**, 237.
- 21 P. de Gennes, *Eur. Phys. J. E: Soft Matter Biol. Phys.*, 2000, **2**, 201.
- 22 T. Salez, J. Salez, K. Dalnoki-Veress, E. Raphaël and J. A. Forrest, *Proc. Natl. Acad. Sci. U. S. A.*, 2015, **112**, 8227.
- 23 R. P. White and J. E. Lipson, *Macromolecules*, 2021, **54**, 4136.
- 24 S. Mirigian and K. S. Schweizer, *J. Chem. Phys.*, 2015, **143**, 244705.
- 25 A. Ghanekarade, A. D. Phan, K. S. Schweizer and D. S. Simmons, *Nat. Phys.*, 2023, **19**, 800.
- 26 C. M. Evans, H. Deng, W. F. Jager and J. M. Torkelson, *Macromolecules*, 2013, **46**, 6091.
- 27 R. Inoue, K. Kawashima, K. Matsui, T. Kanaya, K. Nishida, G. Matsuba and M. Hino, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2011, **83**, 021801.
- 28 T. Kanaya, H. Ogawa, M. Kishimoto, R. Inoue, A. Suter and T. Prokscha, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2015, **92**, 022604.
- 29 T. Koga, N. Jiang, P. Gin, M. K. Endoh, S. Narayanan, L. B. Lurio and S. K. Sinha, *Phys. Rev. Lett.*, 2011, **107**, 225901.
- 30 R. D. Priestley, D. Cangialosi and S. Napolitano, *J. Non-Cryst. Solids*, 2015, **407**, 288–295.
- 31 M. Ediger and J. Forrest, *Macromolecules*, 2014, **47**, 471–478.
- 32 V. M. Boucher, D. Cangialosi, H. Yin, A. Schönhals, A. Alegria and J. Colmenero, *Soft Matter*, 2012, **8**, 5119–5122.
- 33 D. Cangialosi, V. M. Boucher, A. Alegria and J. Colmenero, *J. Chem. Phys.*, 2011, 135.
- 34 J. P. Garrahan, P. Sollich and C. Toninelli, in *Dynamical Heterogeneities in Glasses, Colloids and Granular Media*, ed. L. Berthier, G. Biroli, J.-P. Bouchaud, L. Cipelletti and W. van Saarloosand, Oxford University Press, 2011.
- 35 D. Chandler and J. P. Garrahan, *Annu. Rev. Phys. Chem.*, 2010, **61**, 191.
- 36 G. H. Fredrickson and H. C. Andersen, *Phys. Rev. Lett.*, 1984, **53**, 1244.
- 37 R. G. Palmer, D. L. Stein, E. Abrahams and P. W. Anderson, *Phys. Rev. Lett.*, 1984, **53**, 958.
- 38 C. Scalliet, B. Guiselin and L. Berthier, *Phys. Rev. X*, 2022, **12**, 041028.
- 39 M. R. Hasyim and K. K. Mandadapu, *arXiv*, 2023, preprint, arXiv:2310.06584, DOI: [10.48550/arXiv.2310.06584](https://doi.org/10.48550/arXiv.2310.06584).
- 40 A. Tahaei, G. Biroli, M. Ozawa, M. Popovic and M. Wyart, *Phys. Rev. X*, 2023, **13**, 031034.
- 41 L.-H. Zhang and C.-H. Lam, *Phys. Rev. B*, 2017, **95**, 184202.
- 42 M. Lulli, C.-S. Lee, H.-Y. Deng, C.-T. Yip and C.-H. Lam, *Phys. Rev. Lett.*, 2020, **124**, 095501.
- 43 M. Lulli, C.-S. Lee, L.-H. Zhang, H.-Y. Deng and C.-H. Lam, *J. Stat. Mech.: Theory Exp.*, 2021, **2021**, 093303.
- 44 C.-S. Lee, M. Lulli, L.-H. Zhang, H.-Y. Deng and C.-H. Lam, *Phys. Rev. Lett.*, 2020, **125**, 265703.
- 45 C.-S. Lee, H.-Y. Deng, C.-T. Yip and C.-H. Lam, *Phys. Rev. E*, 2021, **104**, 024131.
- 46 X.-Y. Gao, H.-Y. Deng, C.-S. Lee, J. You and C.-H. Lam, *Soft Matter*, 2022, **18**, 2211.
- 47 G. Gopinath, C.-S. Lee, X.-Y. Gao, X.-D. An, C.-H. Chan, C.-T. Yip, H.-Y. Deng and C.-H. Lam, *Phys. Rev. Lett.*, 2022, **129**, 168002.
- 48 X.-Y. Gao, C.-Y. Ong, C.-S. Lee, C.-T. Yip, H.-Y. Deng and C.-H. Lam, *Phys. Rev. B*, 2023, **107**, 174206.
- 49 B. Li, C.-S. Lee, X.-Y. Gao, H.-Y. Deng and C.-H. Lam, *Soft Matter*, 2024, **20**, 1009.
- 50 C.-T. Yip, M. Isobe, C.-H. Chan, S. Ren, K.-P. Wong, Q. Huo, C.-S. Lee, Y.-H. Tsang, Y. Han and C.-H. Lam, *Phys. Rev. Lett.*, 2020, **125**, 258001.
- 51 A. S. Keys, L. O. Hedges, J. P. Garrahan, S. C. Glotzer and D. Chandler, *Phys. Rev. X*, 2011, **1**, 021013.
- 52 C. Herrero and L. Berthier, *arXiv*, 2023, preprint, arXiv:2310.16935, DOI: [10.48550/arXiv.2310.16935](https://doi.org/10.48550/arXiv.2310.16935).
- 53 L. Berthier and G. Biroli, *Rev. Mod. Phys.*, 2011, **83**, 587.
- 54 C.-H. Lam, *J. Stat. Mech.*, 2018, 023301.
- 55 H.-Y. Deng, C.-S. Lee, M. Lulli, L.-H. Zhang and C.-H. Lam, *J. Stat. Mech.*, 2019, 094014.
- 56 M. Mézard, G. Parisi and M.-A. Virasoro, *Spin glass theory and beyond*, World Scientific, 1990.
- 57 D. Long and F. Lequeux, *Eur. Phys. J. E: Soft Matter Biol. Phys.*, 2001, **4**, 371.
- 58 T. Salez, J. Salez, K. Dalnoki-Veress, E. Raphaël and J. A. Forrest, *Proc. Natl. Acad. Sci. U. S. A.*, 2015, **112**, 8227.
- 59 C.-H. Lam and O. K. C. Tsui, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2013, **88**, 042604.